SEAL: Diagnosis-Driven SAR Ship Instance Segmentation

The **GeoField** Unified Geometry-Field Operator — Evidence, Rationale, and Method

SEAL Project — Technical Note

June 21, 2026

Abstract

The inshore accuracy of SAR ship instance segmentation has long been limited, and the dominant response is to **keep adding capacity** (stronger backbones, orientation-aware queries, foundation-model priors, diffusion decoders) — yet a more basic question is rarely answered: for a given strong baseline, *where* does inshore fail — are ships not found, or are found ships poorly masked? This work follows a **diagnose-before-fix** discipline. We first use a *capacity-independent*, AP-additive oracle to decompose the inshore AP_{75} deficit into a **mask-quality** half and a **detection-coverage** half, and a size-resolved analysis localizes the *medium-object shape/boundary* deficit as the largest recoverable pool that resolution-style remedies cannot reach. A leakage audit then shows that the evaluation split can *distort the diagnosis itself*. Guided by this diagnosis, we **borrow** the principles of several proven mechanisms (salience reweighting, shape-adaptive sampling, absolute position, inter-instance relation) and **redesign**, from first principles, a single native operator, **GeoField**: each pixel computes *one* shared geometry code from which an oriented sampling kernel, a salience that is *structurally inseparable* from the sampling, and a position phase are *simultaneously* derived — mathematically entangled, not stacked. It **enriches** the shared mask feature rather than competing with the readout in logit space, so it cannot regress the way a rigid prior did. Throughout, an **equal-parameter control** separates "mechanism gain" from "raw capacity." This note gives the full evidence chain and rationale, with a framework diagram, algorithm pseudocode, and the equal-parameter ladder.

1 Problem and Motivation

SAR ship instance segmentation is evaluated chiefly on PSeg-SSDD and HRSID with an inshore/offshore stratification (the field's de-facto standard). Inshore, ships abut quays, cranes, and other man-made metallic clutter, and accuracy collapses. The two strongest current methods — the diffusion-based DiffSARShipInst and the orientation-enhanced O^2Former — are **both built on ResNet-50 with a Mask2Former-style decoder**; each *adds capacity* and reports aggregate or stratified AP, but **none** performs an attribution of *where* the inshore deficit arises. Our carrier (denoted A0) is exactly Mask2Former-R50, the recognized field baseline.

2 Core Idea: Diagnose Before Fix

We hold one methodological principle: **the diagnosis is the protagonist; a mechanism is a falsifiable corollary of the diagnosis.** Without first quantifying *which axis* the deficit lies on, no architectural gain can be attributed to the mechanism it claims, and effort may target a sub-problem that is not the binding constraint. The full logical chain is shown in Fig. 1; each step takes the unique exit forced by the previous one as its entry.

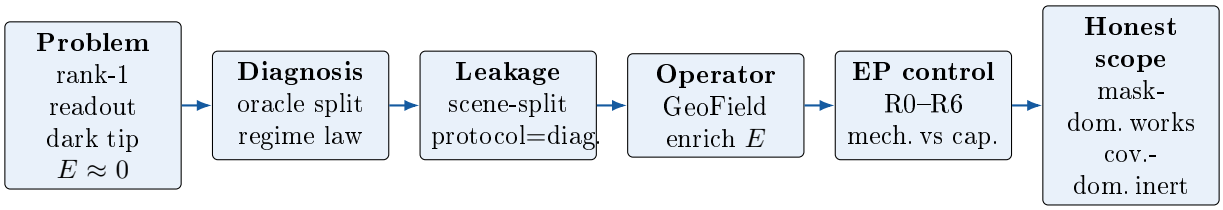


Figure 1: SEAL's argumentative spine: diagnosis-driven, equal-parameter-controlled, scope-honest.

3 Diagnostic Evidence

(1) AP-additive oracle decomposition (capacity-independent). On the A0 carrier, holding detection fixed and substituting *literal ground-truth masks*, we re-run COCO evaluation to obtain the recoverable *ceiling* of inshore AP_{75}. Result: the inshore deficit is **about half mask-quality and half detection-coverage** — not "mask is the sole bottleneck," but a **regime-dependent two-way constraint**.

(2) Regime law (five points). Tabulating base AP_{75}, mask share, and coverage share by dataset/stratum (Table 1): lower base ⇒ more coverage-dominated; mask share falls with difficulty.

Table 1: The regime-dependent two-way constraint (mask% / coverage%).

Stratum	base AP_{75}	mask%	coverage%
PSeg inshore	0.341	51	49
PSeg all	0.615	69	31
PSeg offshore	0.735	87	13
HRSID official (leak)	0.378	43	57
HRSID scene (leak-free)	0.206	**23**	**77**

(3) Size-resolved decomposition — locating the target. Stratifying the mask-quality oracle by object size gives the recoverable AP_{75}:

$$\text{small} \quad +0.253, \qquad \boxed{\text{medium} \quad +0.394} \text{ (largest)}, \qquad \text{large} \quad +0.000 \text{ (saturated)}. \tag{1}$$

The **medium-object recoverable pool is the largest**; yet the resolution-style refinement head below *only reaches small* (a small object spans ∼5 mask pixels at 1/4 stride — it is resolution-limited) and is **inert on medium**. The medium deficit is therefore one of *shape/boundary detail* and needs a different lever — precisely GeoField's target axis.

(4) Leakage audit — protocol welded to diagnosis. The official HRSID split has heavy chip leakage; the leak-free *scene-level* split is the truth. Leakage not only inflates AP but **disguises "coverage-dominated" (23% mask) as "balanced" (43% mask)** — *without fixing leakage, one even mis-judges which axis binds.* All cross-dataset claims must therefore run on the leak-free scene-split.

4 Negative Results and Lessons (keep the essence, discard the dross)

The value of the diagnostic discipline is that it **rejected several of our own well-motivated designs** on the spot — these NO-GOs are the basis for the subsequent design and the collateral for the paper's credibility.

- ShapePrior / GeoKernel (NO-GO): a gated rigid oriented-ellipse prior added in *logit space*, $M_q = \sigma(e_q \cdot E + g\,P)$. Across three seeds it **regressed the baseline** (geo 0.597 < blind 0.609 < baseline

0.638), and geo did not beat blind. Root cause: **a rigid prior fights the optimized rank-1 readout**. ⇒ Lesson: enrich, do not compete.

- DAOM-Seed (inert): a training-time one-to-many auxiliary seeding was indistinguishable from its content-matched null, and both regressed. ⇒ Lesson: a relational/auxiliary mechanism must beat its own null to count.
- Refinement head (PASS, honestly attributed): PSeg overall AP_{75} +3.0 **(3-seed mean)**; an *equal-parameter control* proves the gain is **capacity, not resolution** (caps=refine= 0.643 ≫ bilinear 0.628). It recovers small and is inert on medium. ⇒ Lesson: an equal-parameter control can refute the authors' own narrative.

5 Borrow and Redesign

We **do not stack modules**. We distill several proven mechanisms into *principles* and redesign them, from first principles, into *one* operator (Table 2). Its lineage traces to a self-validated semantic-segmentation teacher (UNet++, with a CoordConv mask head, dynamic snake convolution, CBAM, BatchFormer), but **one cannot point to those modules in the graph** — there is only a bottleneck, a shared code, and a set of free bases.

Table 2: Four principles → read-outs of one shared computation (why it is a redesign, not a stack).

Principle (lineage)	Form inside GeoField	Inseparability
CBAM → salience	$s_p = \sigma(a\, e_p + \rho_p)$, $e_p = \lVert E_{\text{mix}} - E \rVert$	**no own projection — to ablate s you must remove kernel k**
DSConv → shape sampling	per-pixel oriented kernel k_p from a free low-rank basis $\{B^m\}$	predicted weights, not offset-gather; not Gaussian/harmonic
CoordConv → position	coords enter the bottleneck before the code, plus phase ρ_p	into the feature, not the logit
BatchFormer → relation	*training-only* RKD relational regularizer on the code	zero inference params, batch-independent

6 GeoField: a Unified Geometry-Field Operator

6.1 Landing site and non-regression

A Mask2Former mask is the rank-1 readout $M_q = \sigma(e_q \cdot E)$: each query vector e_q is dotted with a *shared* per-pixel feature E, with **no per-instance spatial degree of freedom**, so the low-backscatter dark bow/stern ($E \approx 0$) is systematically under-segmented. GeoField **enriches** that shared feature:

$$E' = E + \gamma \cdot s_p \odot E_{\text{mix}}, \tag{2}$$

feeding the *unchanged* same readout $\langle e_q, E' \rangle$ (detection and segmentation share E', ensuring coherence). It **never adds to logit space**, and the residual gate γ is LayerScale-initialized to 0, so at step 0 it is *byte-identical in output* to the baseline refinement head — structurally it cannot repeat the rigid prior's regression / numerical collapse.

6.2 Framework and internal structure

Fig. 2 shows the overall framework and where GeoField sits (Plan A head-slot; Plan B threaded into the pixel decoder). Fig. 3 shows the operator internally — "one shared code → three entangled read-outs."

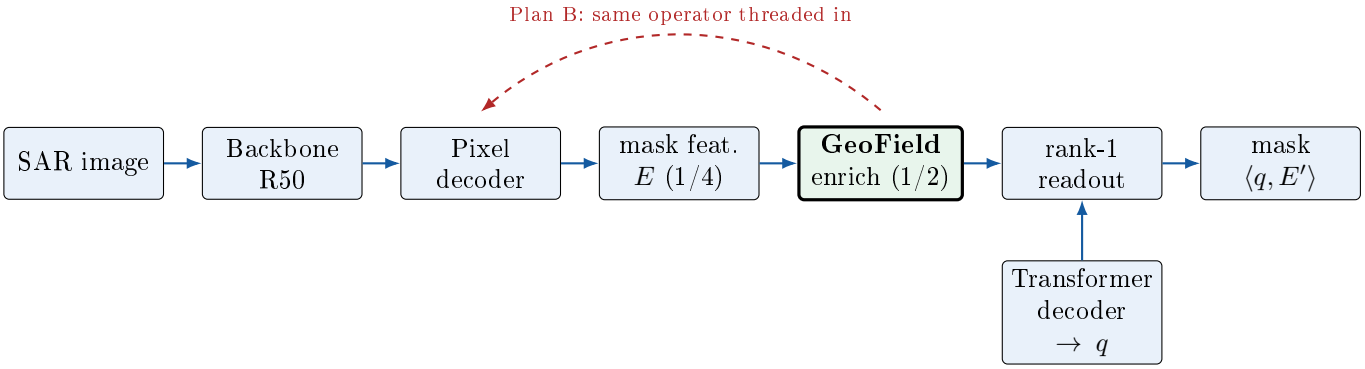


Figure 2: Overall framework. GeoField enriches the shared feature E in the head-slot (Plan A, solid); Plan B threads the *same* operator into the pixel-decoder input (dashed), at the same depth as the field's network-level co-adaptation.

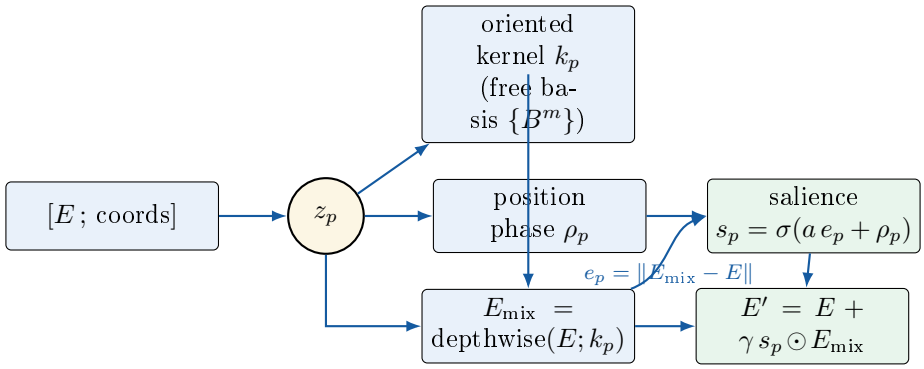


Figure 3: Inside GeoField: a single shared code z_p simultaneously yields the oriented sampling kernel, the position phase, and the salience; salience is set by the kernel-mix residual energy e_p and is thus **structurally inseparable** from the sampling.

6.3 Algorithm (pseudocode)

7 Equal-Parameter Ladder

One class, one `gen_mode` switch, **byte-identical parameters** (21,698,148 per arm, measured); the switch only changes the forward computation (zero the coordinates, freeze the kernel to isotropic, degrade salience to a content gate) and *adds/removes no parameters.*

Why medium is the target: (i) the oracle says the medium recoverable mask pool is the largest (+0.394); (ii) the refinement head is inert on medium — a *shape*, not resolution, problem, exactly what GeoField attacks; (iii) **capacity is inert on medium** (in the pilot, R1≈R0), so any R5 gain on medium is *attributable to the mechanism alone* — the equal-parameter control has its sharpest discriminating power on this axis.

8 Co-Adaptation (more than the head)

Field SOTA all **modify the baseline network**, not just the mask head: O^{2}Former co-adapts at the *feature input* (direction-aware convolution before the pixel decoder) and *query initialization*; DiffSARShipInst modifies the neck/loss/mask-reconstruction. Therefore:

- **Plan A (current)**: GeoField enriches the refined E in the head-slot, for a *cheap, controlled* mechanism test.
- **Plan B (after GO)**: thread the *same* operator into the **pixel-decoder input** — the diagnosis says the $E \approx 0$ dark tip *originates upstream*, so this is the highest-value co-adaptation site, at the

Algorithm 1: GeoField forward (gen_mode selects the ladder arm)

Input : feature $E \in \mathbb{R}^{B\times C\times H\times W}$; mode $\in \{\text{off}, \text{capacity}, \text{pos}, \text{shape}, \text{full}\}$
Output: enriched feature E' (same shape)

if *mode* $=$ *off* **then**
  return E;
coords $\leftarrow$ normalized $(x, y) \in [-1, 1]$; zero them if $\neg$use_pos;
$z \leftarrow W_1\,\mathrm{SiLU}(\mathrm{GN}(W_0[E; \text{coords}]))$; `// the one shared geometry code (1×1 bottleneck)`
if *use_shape* **then**
  $\alpha \leftarrow W_k z$; $k \leftarrow \mathrm{softmax}_{ij}\big(\sum_m \alpha_m B^m\big)$; `// free-basis oriented kernel, fp32`
else
  $k \leftarrow$ isotropic uniform kernel $\frac{1}{K^2}$; `// shape off`
$E_{\text{mix}} \leftarrow \sum_{i,j} k_{ij} \cdot \text{shift}_{ij}(E)$; `//` $K \times K$ `shift-accumulate, memory` $O(BCHW)$
$\rho \leftarrow \tanh(W_{\text{pos}} z)$;
if *use_shape* **then**
  $e_p \leftarrow \|E_{\text{mix}} - E\|_2$; $s \leftarrow \sigma(a\, e_p + \rho)$; `// salience ≡ kernel energy`
else
  $s \leftarrow \sigma(\rho)$;
return $E + \gamma\, s \odot E_{\text{mix}}$; `//` γ `init 0 ⇒ identity`

Table 3: Equal-parameter ladder and decision criteria (target = the medium stratum).

Arm	Name	What it isolates / criterion
R0	off	bypass = stock refinement head (sanity)
R1	capacity	pure capacity (constant coords, isotropic kernel, free content gate)
R2	pos	+ position (answers "is the sine PE already enough?")
R3	shape	+ oriented free-basis kernel + shape-coupled salience
R5	full	the complete operator (note: **no R4 salience-only by construction** = novelty evidence)
R6	full+rel	+ training-only RKD relation (inference params identical to R5)

Criteria: fusion>capacity ⇔ R5>R1; entanglement ⇔ R5> max(R2,R3); relation ⇔ R6>R5 (all over multi-seed boots

same depth as O^{2}Former, but *unifying* its "fixed-angle rotation branches + polar encoding" with a single shared-code free-basis kernel.

9 Current Evidence (verification status)

10 Story and Positioning

SEAL **shares the same baseline** (Mask2Former-R50) with the two 2025 SOTA methods, but pursues a *diagnosis-driven, equal-parameter-controlled, scope-honest* co-adaptation rather than SOTA-chasing. The key narrative is that **the diagnosis is the protagonist**: whether the GeoField pilot is a strong win, a capacity result, or a tie, the paper stands — a strong win is "diagnosis → targeted design → mechanism gain attributed by the equal-parameter control"; a tie is "even a diagnosis-targeted operator cannot beat the control ⇒ it strengthens the two-way-constraint thesis." The predicted inertness on the coverage-dominated HRSID is the diagnosis *correctly predicting* that mask-side help should not transfer, reported as a *declared negative-transfer control*. We do **not** claim to beat competitors on raw AP at 14× fewer epochs; the contribution is the *diagnostic methodology*

Table 4: Completed verifications (measured).

Item	Result	Status
Operator CPU smoke	equal-param / residual-identity / finite fwd+bwd / AMP-stable / **batch-independent** (2.4×10^{-7}) / arm-differentiation	7/7 pass
Head-level equal-param	all six arms 21,698,148 params (byte-identical)	pass
GPU training smoke	R5, real data, 12 iters: loss monotonically down, no NaN/OOM, 8.6 GB	pass
Deployment verification	ONNX export / torch-ONNX-TensorRT parity (4.8×10^{-7}) / batch-independent	pass
Deployment FPS	reference impl is memory-bound (fp32 $\approx$ 1.5$\times$ a plain conv); fused kernel is future work	honestly logged
4-arm pilot (PSeg)	R0/R1 done (medium mAP 0.444/0.441, capacity inert on medium); R3/R5 in progress	in progress

$+$ *the unified operator* $+$ *the leakage audit.*

Note: all numbers are verified measured/oracle values; the pilot arms R3/R5 are still training, and the verdict will be updated once available.